

Ruize Xu

Homepage: <https://rick-xu315.github.io>

Email : ruize.xu.gr@dartmouth.edu

Mobile : +1-917-407-9605

EDUCATION EXPERIENCE

- **Dartmouth College, NH** Sep. 2025 - May 2030 (expected)
PhD in Computer Science
Advisor: [Nikhil Singh](#)
- **Columbia University, NY** Jan. 2024 - May 2025
MS in Computer Science-Thesis Track
GPA: 4.08/4.0, Advisor: [Zhou Yu](#)
- **Renmin University, China (Ranked 20th globally on [CSRankings 2025](#))** Sep. 2019 - Jun. 2023
BS in Data Science
- **University of California, Davis, CA** Jan. 2022 - Apr. 2022
Exchange Student

RESEARCH INTEREST

Post-training Dynamics of Large Language Models:

RL for Specialization, Policy Optimization, World Model Learning

Mechanistic and Representation-level Interpretability:

Geometric Analysis of LLM Representations and Structure Emergence, Causal and Data-centric Interpretation

RESEARCH EXPERIENCE

- **Columbia University** Feb 2024 - Present
 - Mentor: [Junfeng Yang](#)
 - Project: Adversarial analysis of diffusion-based image detectors via internal guidance (CVPR 2024 Workshop).
 - Mentor: [Zhou Yu](#)
 - Project: Built resume-job matching framework with hard-negative mining and synthetic data generation (ACL 2025).
- **Alibaba Security Group** Jul 2023 - Dec 2023
 - Mentor: [Pengda Qin](#)
 - Projects: Built supervised fine-tuning data for 7B/14B LVLMS for multi-image QA (MME 1400+). Expanded LVLMS with MoE to improve unseen-task generalization (+50%) while maintaining prior performance (95%).
- **Baidu Vision** Nov 2022 - Jun 2023
 - Mentor: [Yumeng Zhang](#)
 - Projects: Improved NeRF for dynamic unbounded autonomous-driving scenes via tri-plane decomposition on NuScenes.
- **Renmin University of China (Undergraduate Research)** Jun 2021 - Jun 2024
 - Advisor: [Di Hu](#)
 - Projects: Studied modality imbalance in multi-modal learning; proposed modality-wise normalization to improve weak uni-modalities in fine-grained audio-visual tasks (ICASSP 2023).

PUBLICATION LIST

1. **How do World Models and Policies Compose in LLM Agents? A Joint Spectral and Behavioral Account.**
Ruize Xu, Xiao Yu, Yujin Tang, Chenzhuang Shang, Nikhil Singh.
Findings of EMNLP 2026. ([Paper](#))
2. **OpenForgeRL: Train Harness-native Agents in Any Environment.**
Xiao Yu, Baolin Peng, Ruize Xu, Han Zou, Qingyun Wu, Hao Cheng, Weiran Yao, Nikhil Singh, Zhou Yu, Jianfeng Gao.
arXiv preprint, 2026. ([Paper](#))
3. **DMV-Bench: Diagnosing Long-Horizon Multimodal Agents' Visual Memory with Incidental Cue Injection.**
Yujin Tang, Chenzhuang Shang, Ruize Xu, Nikhil Singh.
arXiv preprint, 2026. ([Paper](#)) ([Code](#))

4. **ConFit v3: Improving Resume-Job Matching with LLM-based Re-Ranking.**
Xiao Yu*, **Ruize Xu***, Chengyuan Xue*, Junchen Chen, Matthew So, Shiqi Ma, Bing Liu, Xiaodan Liang, Zhou Yu.
arXiv preprint, 2026. ([Paper](#))
5. **Reinforcement World Model Learning for LLM-based Agents.**
Xiao Yu, Baolin Peng, **Ruize Xu**, Yelong Shen, Pengcheng He, Suman Nath, Nikhil Singh, Jianfeng Gao, Zhou Yu.
arXiv preprint, 2026. ([Paper](#))
6. **AI Agents for Web Testing: A Case Study in the Wild.**
Naimeng Ye*, Xiao Yu*, **Ruize Xu***, Tianyi Peng, Zhou Yu.
NeurIPS 2025, LAW Workshop. ([Paper](#))
7. **Dyna-Think: Synergizing Reasoning, Acting, and World Model Simulation in AI Agents.**
Xiao Yu, Baolin Peng, **Ruize Xu**, Michel Galley, Hao Cheng, Suman Nath, Jianfeng Gao, Zhou Yu.
ICLR 2026, Workshop on World Models: Understanding, Modelling and Scaling. ([Paper](#))
8. **ConFit v2: Improving Resume-Job Matching using Hypothetical Resume Embedding and Runner-Up Hard-Negative Mining.**
Xiao Yu*, **Ruize Xu***, Chengyuan Xue*, Jinzhong Zhang, Xu Ma, Zhou Yu.
Findings of ACL 2025. ([Paper](#))
9. **ComCLIP: Training-Free Compositional Image and Text Matching.**
Kenan Jiang, Xuehai He, **Ruize Xu**, Xin Eric Wang.
NAACL 2024. ([Paper](#))
10. **From Detection to Deception: Are AI-Generated Image Detectors Adversarially Robust?**
Yun-yun Tsai, **Ruize Xu**, Chengzhi Mao, Junfeng Yang.
CVPR 2024, Responsible Generative AI Workshop. ([Paper](#))
11. **MMCosine: Multi-Modal Cosine Loss Towards Balanced Audio-Visual Fine-Grained Learning.**
Ruize Xu, Ruoxuan Feng, Shi-Xiong Zhang, Di Hu.
ICASSP 2023. ([Paper](#)) ([Code](#))

HONORS AND AWARDS

- Silver Reviewer of ICML. 2026
- Tinker Research Grant **\$5,000**, *Thinking Machines Lab*. 2026
- Renmin University of China (RUC): National 1st Prize, Chinese Mathematics Competitions; Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling. Meritorious Winner (Top 7%) 1st Prize Scholarship for Academic Excellence (Top 5%); President Scholarship for Exchange Students; Undergraduate Research Fund; Dean's MingDe Data Science Talents Nomination (17/321); Academic Excellence Award (Top 3% GPA) 2020–2022

SERVICE

- **Reviewer:** [ICASSP 2024](#), [Neurips LAW Workshop 2025](#), [ICLR 2026 World Models Workshop](#), [ICML 2026](#), [NeurIPS 2026](#)
- Table Tennis School Team: RUC&Columbia